International Cybersecurity and Digital Forensics Academy (ICDFA)

GRC205: Incident Management and Business Continuity (Week 21)

BUSINESS CONTINUITY AND

DISASTER RECOVERY PLAN

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Scenario organisation	Northbridge Skills Institute
Scenario	Day 0, 08:00 WAT main site outage (72-hour flooding / power failure)
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Executive Mandate and Scope

At 08:00 WAT on Day 0 (Monday), flooding and a severe power failure put Northbridge Skills Institute's main site out of action for an estimated 72 hours. LMS01, FIN01, REG01, HR01, FS01 and AUTH01 are inaccessible.

This plan follows ISO 22301:2019 and NIST SP 800-34 Rev. 1. It covers a five-function business impact analysis (BIA), dependency analysis, warm site recovery scheduling and a tabletop test, so critical services return within their recovery time objectives (RTOs) without losing data integrity.

Task 1: Business Impact Analysis

1.1 Five-Function BIA Matrix

Function and systems	Owner and users	Dependencies	Impact and constraints	Minimum acceptable service	Priority
F1: Teaching and Assessment (LMS01, DB01)	Registrar (1,800 learners, 90 faculty)	AUTH01, DB01, network, M365, laptops/BYOD	11:00 WAT exam for 120 learners. NGN 120k reschedule fee if missed. Severe academic loss.	Online exam portal; 15 centre PCs + 80 BYOD spots; paper attendance roster.	**1 (Critical)**
F2: Fees and Finance (FIN01, DB01)	Finance Lead (CFO, accounts, payers)	AUTH01, DB01, network, external gateway	16:00 WAT Day 0 fee confirmation deadline. Payment provider is live but the local ledger is not updated.	Offline paper receipt log; payment gateway web portal checks; manual batch entry.	**2 (High)**
F3: Admissions and Registration (REG01, DB01)	Admissions Head (new applicants)	AUTH01, DB01, network, M365 email	18:00 WAT Day 0 admissions close. New applications cannot enter the local database.	M365 shared spreadsheet for intake; manual timestamping.	**3 (Medium-High)**
F4: Staff Payroll (HR01, DB01)	HR Manager (90 staff)	AUTH01, DB01, network, bank portal	12:00 WAT Day 1 bank submission. Payment delays hurt staff welfare and trust.	Manual payroll check using the pre-disruption exported sheet.	**4 (Medium)**
F5: Records and Certificates (FS01)	Records Lead (graduates, third parties)	AUTH01, FS01, network, storage	48-hour internal service commitment. Certificate requests delayed; low immediate loss.	Urgent requests logged by M365 email; manual verification once FS01 is restored.	**5 (Low)**

1.2 Disruption Cost (8-Hour and 24-Hour Scenarios)

Costs use the supplied hourly NGN rates plus the one-off NGN 120,000 exam rescheduling fee. They assume no continuity measures from the 08:00 WAT baseline.

Function	Hourly rate (NGN)	8-hour cost	24-hour cost	Calculation
F1: Teaching and Assessment	60,000	NGN 600,000	NGN 1,560,000	(8h × 60k) + 120k = 600k; (24h × 60k) + 120k = 1.56M
F2: Fees and Finance	40,000	NGN 320,000	NGN 960,000	8h × 40k; 24h × 40k. 16:00 fee deadline breached.

Function	Hourly rate (NGN)	8-hour cost	24-hour cost	Calculation
F3: Admissions and Registration	20,000	NGN 160,000	NGN 480,000	8h × 20k; 24h × 20k. 18:00 deadline breached.
F4: Staff Payroll	10,000	NGN 80,000	NGN 240,000	8h × 10k; 24h × 10k. Day 1 12:00 bank deadline approached.
F5: Records and Certificates	5,000	NGN 40,000	NGN 120,000	8h × 5k; 24h × 5k. Within the 48h commitment.
Total	**135,000**	**NGN 1,200,000**	**NGN 3,360,000**	**Combined 8h = NGN 1.20M; 24h = NGN 3.36M**

Limits of the financial estimates

Limitation	Effect
Non-linear cost spikes	The model treats cost as linear, but real impact jumps when a regulatory or contractual deadline passes.
Qualitative costs left out	Candidate anxiety, brand damage, regulatory scrutiny and possible loss of accreditation are not captured.

Missing the 11:00 WAT exam harms assessment integrity and the welfare of 1,800 learners. The reputational loss far exceeds the NGN 120,000 fee.

1.3 Priority Sequence and Technical Dependencies

Business priority follows deadlines and impact. Technical restoration follows infrastructure dependencies. F1 is the top priority because of the 11:00 WAT exam, but LMS01 cannot start until the recovery network, AUTH01 and DB01 are up.

Restoration stage	Duration	Depends on
1. Recovery network	30 min	None
2. AUTH01 directory	30 min	Recovery network
3. Security and backup checks	30 min	AUTH01
4. DB01 database	90 min	Security checks
5. Applications: LMS01 / FIN01 / REG01 / HR01	45 / 30 / 30 / 45 min	DB01
FS01 file server	60 min	AUTH01 and security checks

Three critical dependency risks

Risk	Why it matters
Single point of failure: DB01	It hosts all four application databases.
Shared AUTH01 directory	If it fails, every system login is blocked.
Single network path to the warm site	One link failure cuts off the whole recovery site.

Task 2: Recovery Objectives and Backup Design

2.1 Recovery Objective Matrix

Each function has a safety buffer between its RTO and its maximum tolerable downtime (MTD).

Function	Target RTO	Target RPO	MTD	Buffer	Declared service level
F1: Teaching and Assessment	4 hours (12:00 WAT)	1 hour (07:00 WAT)	6 hours (14:00 WAT)	2 hours	Core exam portal restored at the warm site; BYOD + 15 centre PCs live.
F2: Fees and Finance	6 hours (14:00 WAT)	2 hours (06:00 WAT)	10 hours (18:00 WAT)	4 hours	FIN01 online before the 16:00 fee deadline; transaction logs reconciled.
F3: Admissions and Registration	8 hours (16:00 WAT)	4 hours (04:00 WAT)	12 hours (20:00 WAT)	4 hours	REG01 online before 18:00 admissions close; M365 intake synced.
F4: Staff Payroll	18 hours (02:00 Day 1)	12 hours (20:00 Day -1)	28 hours (12:00 Day 1)	10 hours	HR01 online well before the Day 1 12:00 bank deadline.
F5: Records and Certificates	24 hours (08:00 Day 1)	24 hours (08:00 Day -1)	48 hours (08:00 Day 2)	24 hours	FS01 file shares restored within the 48h commitment.

2.2 Backup Checkpoint Analysis

At the 08:00 WAT disruption, the offsite catalogue held four recovery items. The table compares each item's age with its target RPO.

Backup item	Checkpoint (WAT)	Age at disruption	Target RPO	Data loss window	Feasibility and gap
B01: DB01 full backup	23:00 Day -1	9 hours	1–4 hours	9 hours (if B02 fails)	Unacceptable for F1/F2. Heavy manual re-entry if the incremental fails.
B02: DB01 incremental	07:00 Day 0	1 hour	1 hour (F1)	1 hour (07:00–08:00)	Meets F1/F2 RPO. One hour of transactions reconciled via gateway logs.
B03: AUTH01 snapshot	00:00 Day 0	8 hours	12 hours	8 hours (00:00–08:00)	Meets RPO. Directory changes made 00:00–08:00 are re-entered.
B04: FS01 full backup	23:00 Day -1	9 hours	24 hours (F5)	9 hours (23:00–08:00)	Fully meets the 24h RPO. Files created after 23:00 are re-uploaded from email.

2.3 Backup Schedule and Verification

To close the RPO gaps and remove single-point-of-failure administration, the plan uses a 3-2-1-1-0 design: 3 copies, 2 media types, 1 offsite, 1 offline/immutable, and 0 unverified restore errors.

Element	Design
Schedule	DB01 transaction log backups hourly (08:00–22:00), daily full backup at 23:00, and 15-minute log shipping to an isolated cloud vault.
Restoration chain	B01 (full) plus all sequential transaction logs up to the chosen recovery point.

Element	Design
Automated sandbox verification	Offsite replication alone does not prove recoverability. Sandbox restore tests run weekly, checking database checksums, table consistency and access independence before a backup is marked ready.

Task 3: Recovery Strategy and Budget

3.1 Strategy Comparison (Cold vs Warm vs Hot Site)

Package	First critical service	Data-loss window	Staffing and technical fit	Malware / propagation risk	RTO/RPO alignment
Cold site	24–48 hours	Up to 24 hours	Exceeds the 2-technician capacity; heavy bare-metal build.	Low; clean hardware rebuild.	**Fails F1, F2, F3 RTOs.** Unacceptable for the 11:00 exam.
Warm site	3–5 hours	1–6 hours	Fits 2 technicians; pre-configured hardware ready.	Medium; reduced by sandbox EDR scanning.	**Meets all RTO/RPO targets** with minor business workarounds.
Hot site	About 1 hour	Up to 15 minutes	Needs dedicated cloud ops staff; high complexity.	High; real-time sync can spread malware.	Exceeds targets but uses 100% of the budget.

3.2 Annual Budget Analysis

Each package is tested against the NGN 3,000,000 annual limit, plus mandatory NGN 400,000 for protected backups and NGN 200,000 for training and drills.

Package	Base cost (NGN)	Protected backups	Training and drills	Total annual cost	Variance vs NGN 3.0M cap
Cold site	600,000	400,000	200,000	**NGN 1,200,000**	+NGN 1,800,000 (under budget, capability deficit)
Warm site	1,500,000	400,000	200,000	**NGN 2,100,000**	**+NGN 900,000 (recommended; optimal buffer)**
Hot site	2,400,000	400,000	200,000	**NGN 3,000,000**	NGN 0 (at limit, no contingency)

Recommendation and return on investment

Item	Justification
Recommendation	Warm site package at NGN 2,100,000 a year. It balances capability, feasibility for a 2-technician team and cost, and leaves a NGN 900,000 contingency buffer.
ROI	One unmitigated 24-hour outage costs NGN 3,360,000 in direct disruption. At NGN 135,000 an hour, the package pays for itself after about 16 hours of downtime avoided in a year.

Task 4: Business Continuity and Disaster Recovery Plan

4.1 Governance and Activation

The plan is activated when a site outage is confirmed or expected to exceed 4 hours. The Incident Commander (Operations Lead) holds activation authority, supported by the Deputy Commander (IT Lead).

4.2 BC/DR RACI Matrix

R = Responsible; A = Accountable; C = Consulted; I = Informed. Each activity has one A.

Activity	Incident Cmd	IT Recovery	Facilities	Admissions/ Reg	Comms	CFO
1. Plan activation and site declaration	**A/R**	C	C	I	I	I
2. Alternate facility setup (25 PCs)	**A**	R	R	I	I	I
3. Temporary exam execution (BYOD/staggered)	**A**	C	I	R	R	I
4. Technical backup restore and validation	I	**A/R**	I	I	I	I
5. Stakeholder and learner communication	**A**	I	I	C	R	I
6. Primary site demobilisation and failback	**A**	R	R	C	I	C

4.3 Temporary Workaround Procedures (72-Hour Outage)

The alternate facility provides 25 workstations for the 120-learner exam.

Item	Arrangement
BYOD learners	80 learners use personal devices over secure guest Wi-Fi.
Centre learners	40 learners in two staggered 20-learner shifts (Shift 1: 12:00 WAT, Shift 2: 13:30 WAT) on 20 of the 25 workstations. 5 are kept as hot spares.
Academic integrity	Randomised question banks and paper attendance sheets.

4.4 Technical DR Runbook

Step	Action and target system	Prerequisite	Owner	Verification and rollback
1	Bring up the warm site recovery network	Warm site power and gateway active	Tech A	Ping gateway and DNS test. **Rollback:** switch to the backup WAN interface.
2	Restore AUTH01 directory (B03)	Step 1 complete	Tech A	Authenticate a test user. **Rollback:** re-apply snapshot B03.
3	Run security and backup checks	Step 2 complete	Tech A	AV/EDR scan passes; B01/B02 hashes verified. **Rollback:** isolate the network segment.
4	Restore DB01 database (B01 + B02)	Step 3 complete	Tech A	DBCC CHECKDB passes; test table queries. **Rollback:** fall back to B01 full.
5	Restore applications (LMS01 / FIN01)	Step 4 complete	Tech A and B	Login test and query run. **Rollback:** re-mount the app container.

4.5 Stakeholder Communication Matrix and Learner Notice

Stakeholder	Message	Owner	Channel	Frequency
Learners	Exam rescheduling and instructions	Registrar / Comms Lead	SMS and email	At declaration, then every 2 hours
Executive Team	Status, decisions needed, recovery progress	Incident Cmd	Phone / secure messaging	Every 2 hours
Staff	Site closure and working arrangements	Comms Lead	SMS / staff group	Twice daily
Bank	Payroll deadline risk and payment timing	Finance Lead (CFO)	Phone and email	Before the Day 1 12:00 deadline
Regulators	Disruption notice and any data impact	Incident Cmd	Official email	As regulations require

Learner notice (Word count: 102)

IMPORTANT LEARNER NOTICE: ONLINE EXAMINATION RESCHEDULING

TO: All registered examination candidates (120 learners)
FROM: Office of the Registrar, Northbridge Skills Institute | **DATE:** Day 0, 08:30 WAT

Due to a power and flooding disruption at our main site, today's 11:00 WAT examination is **rescheduled** to 12:00 WAT. It will run at our Alternate Facility (Community Tech Centre) and on the BYOD online portal.

What to do:

1. BYOD candidates: log into the LMS portal from 11:45 WAT for a system check.
2. Candidates needing a centre workstation: report to the Alternate Facility by 11:30 WAT. Exams run in two sessions (Shift 1: 12:00 WAT, Shift 2: 13:30 WAT).
3. No candidate will be penalised for delays caused by site transit.

We will send further updates by SMS. Thank you for your cooperation.

Task 5: Recovery Schedule and Exercise

5.1 Warm Site Recovery Schedule (2 Technicians)

The schedule uses the warm site drill durations. Tasks run in parallel only when prerequisites and technician availability allow.

Action	Duration	Prerequisite	Tech	Start (WAT)	Finish (WAT) and status
Recovery network	30 min	None	A	08:00	08:30 (network live)
AUTH01 restoration	30 min	Recovery network	A	08:30	09:00 (auth live)
Security and backup checks	30 min	AUTH01	A	09:00	09:30 (checks pass)
FS01 restoration	60 min	Security checks	B	09:30	10:30 (F5 restored)
DB01 restoration	90 min	Security checks	A	09:30	11:00 (DB live)
FIN01 restoration	30 min	DB01	B	11:00	11:30 (F2 restored)
LMS01 restoration	45 min	DB01	A	11:00	11:45 (F1 restored)

Action	Duration	Prerequisite	Tech	Start (WAT)	Finish (WAT) and status
REG01 restoration	30 min	DB01	B	11:30	12:00 (F3 restored)
HR01 restoration	45 min	DB01	A	11:45	12:30 (F4 restored)

Schedule path analysis

Item	Result
Critical path	Recovery network (30m) → AUTH01 (30m) → security checks (30m) → DB01 (90m) → LMS01 (45m) = 225 minutes (3h 45m). LMS01 goes live at 11:45 WAT.
Against targets	F1 is restored at 11:45 WAT (3h 45m elapsed), inside the 4-hour RTO (12:00 WAT) and 6-hour MTD (14:00 WAT). All five functions are running by 12:30 WAT (4h 30m), within every MTD.

5.2 Tabletop Decision Log (4 Required Injects + 1 Custom)

A 45-minute tabletop exercise tested how well the plan adapts to changing conditions.

Inject	Decision	Owner	Rationale	Communication	Effect on plan
Min 10: Site closure extended to 5 days	Keep the warm site running; extend the alternate facility lease.	Incident Cmd	The primary site is unusable for a week; the warm site handles the load.	Notify Board and staff.	Plan shifts from a 72h sprint to 5 days of sustained operations.
Min 20: Alternate PCs cut from 25 to 15	Move 40 centre learners to two 20-learner shifts (12:00 and 13:30 WAT).	Registrar	15 PCs cannot seat 40 learners at once; staggering fixes the shortfall.	SMS to centre learners.	Shift 2 exam finishes by 15:00 WAT.
Min 30: DB01 B02 fails; unexpected logon	Isolate B02; restore B01 full (23:00 Day -1); start a security logon audit.	IT Recovery Lead	Prevents possible malware spread; falls back to a known clean full backup.	Alert the Security Analyst.	RPO rises to 9h; Day 0 23:00–08:00 data must be re-entered.
Min 40: Payment provider mismatch	Reconcile gateway web API logs against FIN01 after the 11:30 restore.	Finance Lead	The gateway holds the true record; the local DB was offline.	Brief the CFO.	FIN01 ledger updated with zero financial loss by 14:00.
Min 45 (custom): Main WAN link drops at 10:00	Fail over to the redundant 5G cellular gateway at the alternate site.	IT Recovery Lead	Keeps warm site connectivity without delaying the DB01 restore.	Alert the tech team.	No delay to the 11:45 LMS01 finish.

5.3 Corrective Action Plan

Action	Priority	Owner	Target	Measurable evidence
1. Automate transaction log shipping (15-min RPO)	High	IT Recovery Lead	30 days	DB01 log shipping active; automated restore test logs.
2. Expand alternate facility SLA to 40 PCs	High	Facilities Lead	45 days	Signed vendor amendment for 40 guaranteed PCs.
3. Buy dual cellular 5G failover routers	Medium	IT Recovery Lead	14 days	Hardware asset tag and automated failover test report.

Action	Priority	Owner	Target	Measurable evidence
4. Automate payment gateway log reconciliation	Medium	Finance Lead	30 days	Reconciliation script tested and deployed.
5. Hold twice-yearly all-hands BC/DR training	Medium	HR Manager	60 days	85%+ staff attendance records and post-test scores.

Task 6: Governance and Performance Measures

6.1 Governance, Review Triggers and Framework Mapping

The Business Continuity Coordinator owns the plan, with annual sign-off by the Board Audit Committee. Reviews are triggered by major infrastructure changes, post-incident reviews and annual drill findings.

NIST SP 800-34 Rev. 1 step	ISO 22301:2019 clause	Plan deliverable	Climate / flood consideration
Step 2: Business impact analysis	Clause 8.2: BIA and risk assessment	Task 1 BIA matrix and calculations	Tests site vulnerability to a local 100-year flood.
Step 3: Preventive controls	Clause 8.3: Strategies and solutions	Task 2 backup design (3-2-1-1-0)	Offsite backup vault placed outside the local flood plain.
Step 4: Contingency strategies	Clause 8.3: Strategies and solutions	Task 3 strategy and budget	Alternate facility chosen on high ground.
Step 5: Plan development	Clause 8.4: BC plans and procedures	Task 4 BC/DR plan	Plan assumes severe-weather site loss.
Step 6: Testing, training and exercises	Clause 8.5: Exercise programme	Task 5 schedule and tabletop log	Simulates severe weather during the rainy season.

6.2 Key Performance Indicators

KPI	Formula	Data source	Target	Frequency	Owner
RTO attainment rate	(Achieved recovery time ÷ target RTO) × 100	Drill and incident logs	≤ 100%	Semi-annual	IT Recovery Lead
Restoration test coverage	(Systems tested ÷ in-scope systems) × 100	Backup console audit	100% of systems	Quarterly	IT Recovery Lead
CAP closure rate	(Closed actions ÷ total open actions) × 100	BC/DR audit tracker	≥ 90% on time	Quarterly	BC Coordinator
Staff training completion	(Staff completed ÷ total staff) × 100	HR LMS reports	≥ 95% of staff	Annual	HR Manager

6.3 Maturity Assessment and 12-Month Target

Maturity	Position
Current: Level 1.5 (Initial / ad hoc)	Backup restore tests had not run in 90 days, documentation was fragmented and site dependencies were undocumented.
12-month target: Level 3.0 (Defined)	Needs quarterly automated restore drills, contractually guaranteed alternate facility SLAs, and continuous KPI reporting to the Board Audit Committee.

References

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